



NAME:
HABIB ID:

LINEAR ALGEBRA

SPRING 2026

SECTIONS L1, L2

QUIZ 1

Max Marks: 10

Time: 8 minutes

(a) Define the inverse of a matrix A .

[3]

IF Q IS A SQUARE MATRIX
AND A IS ALSO A SQUARE MATRIX,
(BOTH OF ORDER n) THEN Q
IS SAID TO BE AN INVERSE OF
 A IF AND ONLY IF $\rightarrow$ CONVERSE
HOLDS
 $QA = AQ = I_n = I$ AND
IS DENOTED BY A^{-1} i.e. $Q = A^{-1}$

(b) Show that the inverse of any matrix is unique.

[7]

PROOF: $\because B$ IS AN INVERSE OF
 A , WE HAVE $BA = I \rightarrow (1)$
MULTIPLYING BOTH SIDES (OF (1))
ON THE RIGHT BY C GIVES
 $(BA)C = IC = C. \rightarrow (2)$
BUT $(BA)C = B(AC), \because AC = I$
 $= B(I) = B, \rightarrow (3)$
SO THAT $C = B$ FROM
(2) AND (3).
RESULT: INVERSE OF A
NONSINGULAR MATRIX IS
UNIQUE.